DevOps CI/CD Pipeline for Scientific Calculator

Abhishek Prasanna
MT2025007

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Project Overview

- Developed a complete CI/CD pipeline for a Java-based Scientific Calculator
- Automates build, testing, containerization, and deployment
- Triggered automatically through GitHub webhooks
- Ensures fast, reliable, and consistent software delivery

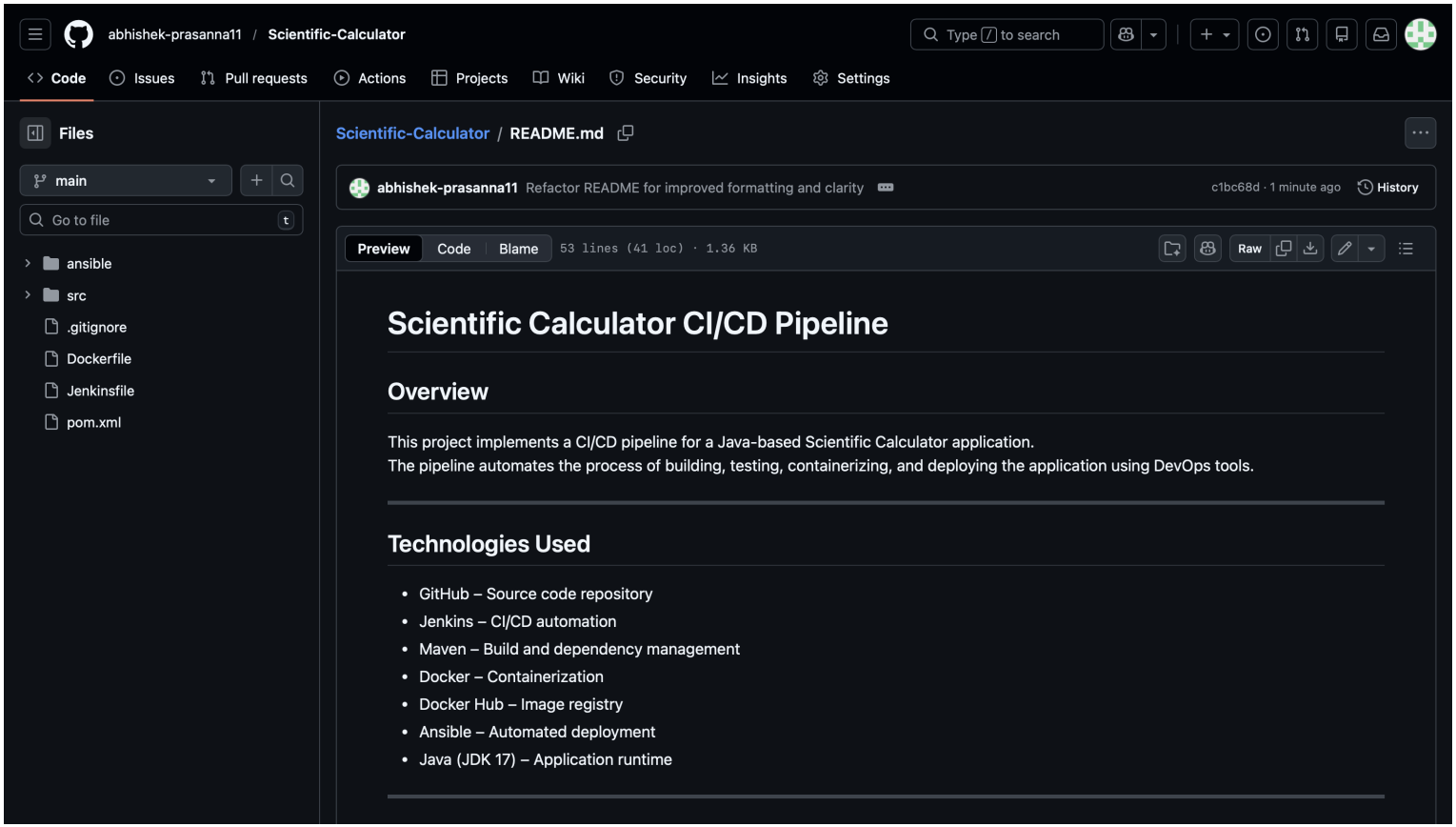
Tools and Technologies

- GitHub – Source code management
- Jenkins – CI/CD automation server
- Maven – Build and dependency management
- Docker – Containerization
- Docker Hub – Image registry
- Ansible – Automated deployment
- Ngrok – Expose Jenkins to internet
- Gmail SMTP – Email notifications

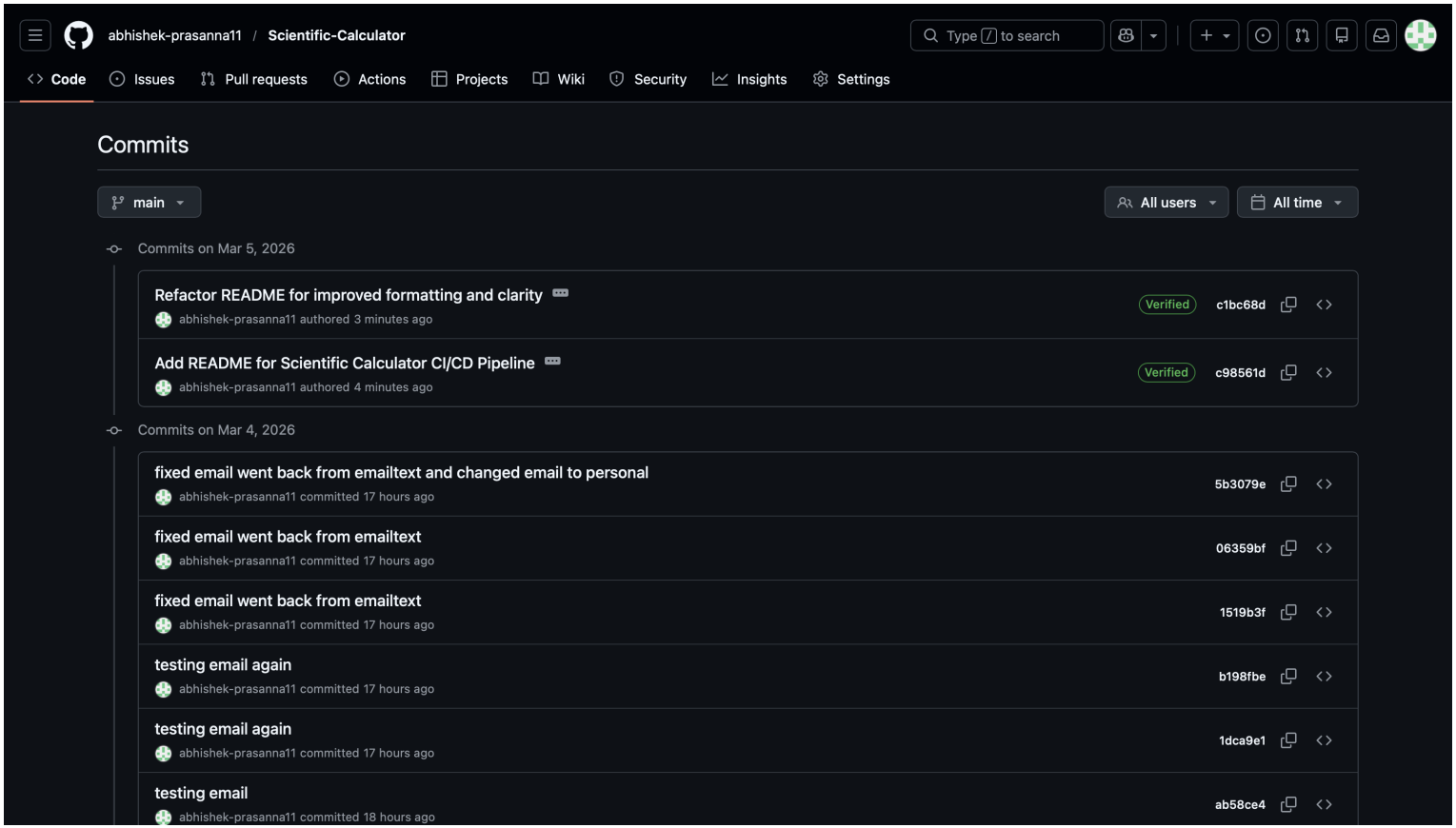
Application Development

- Java-based Scientific Calculator
- Supports arithmetic and scientific operations
- Built using Maven
- Unit testing implemented using JUnit
- Packaged as a JAR file
- The application is containerized using Docker

GitHub Repository



GitHub Commit History



Jenkinsfile Explanation

The Jenkinsfile defines the Continuous Integration and Continuous Deployment (CI/CD) pipeline used in this project. It is a configuration file written using the Jenkins Pipeline Domain Specific Language (DSL) and is stored inside the project repository. By keeping the pipeline definition within the repository, the build process becomes version-controlled, reproducible, and automated.
The pipeline used in this project follows the **Declarative Pipeline model**, which provides a structured and readable syntax for defining pipeline stages and execution steps.
The Jenkins pipeline automates the process of building, testing, packaging, containerizing, and deploying the application.

Jenkinsfile Explanation

Pipeline Structure
A Jenkins pipeline consists of three main components.
Pipeline Block
The pipeline block defines the entire CI/CD workflow and contains all the configuration and stages required for execution.
Agent
The agent directive specifies the execution environment where the pipeline runs. In this project, the pipeline runs on a Jenkins node that has access to Maven and Docker.
Stages
The stages section contains multiple stages that represent different steps of the software delivery pipeline. Each stage performs a specific task such as compiling the application or running tests.

Jenkinsfile Explanation

Checkout Stage

The Checkout stage retrieves the latest source code from the GitHub repository.

- Fetch the latest application source code
- Synchronize the Jenkins workspace with the repository
- Prepare the project files for the build process

Compile Stage

- Ensures that the source code is syntactically correct
- Verifies that all dependencies are available
- Prepares compiled classes for testing

Jenkinsfile Explanation IV

Test Stage

- Executes unit tests using Maven
- Validates application functionality
- Improves reliability of the software

Package Stage

- Creates the application JAR file
- Produces the deployable artifact
- Prepares the application for containerization

Jenkinsfile Explanation V

Docker Build Stage
In this stage, Jenkins builds a Docker image using the Dockerfile present in the repository.
Docker containerization packages the application along with its runtime environment, ensuring that the application behaves consistently across different systems.

- Builds a container image containing the application
- Includes the Java runtime environment
- Provides portability across systems

Jenkinsfile Explanation VI

Docker Push Stage
After the image is built, the Docker image is pushed to Docker Hub.

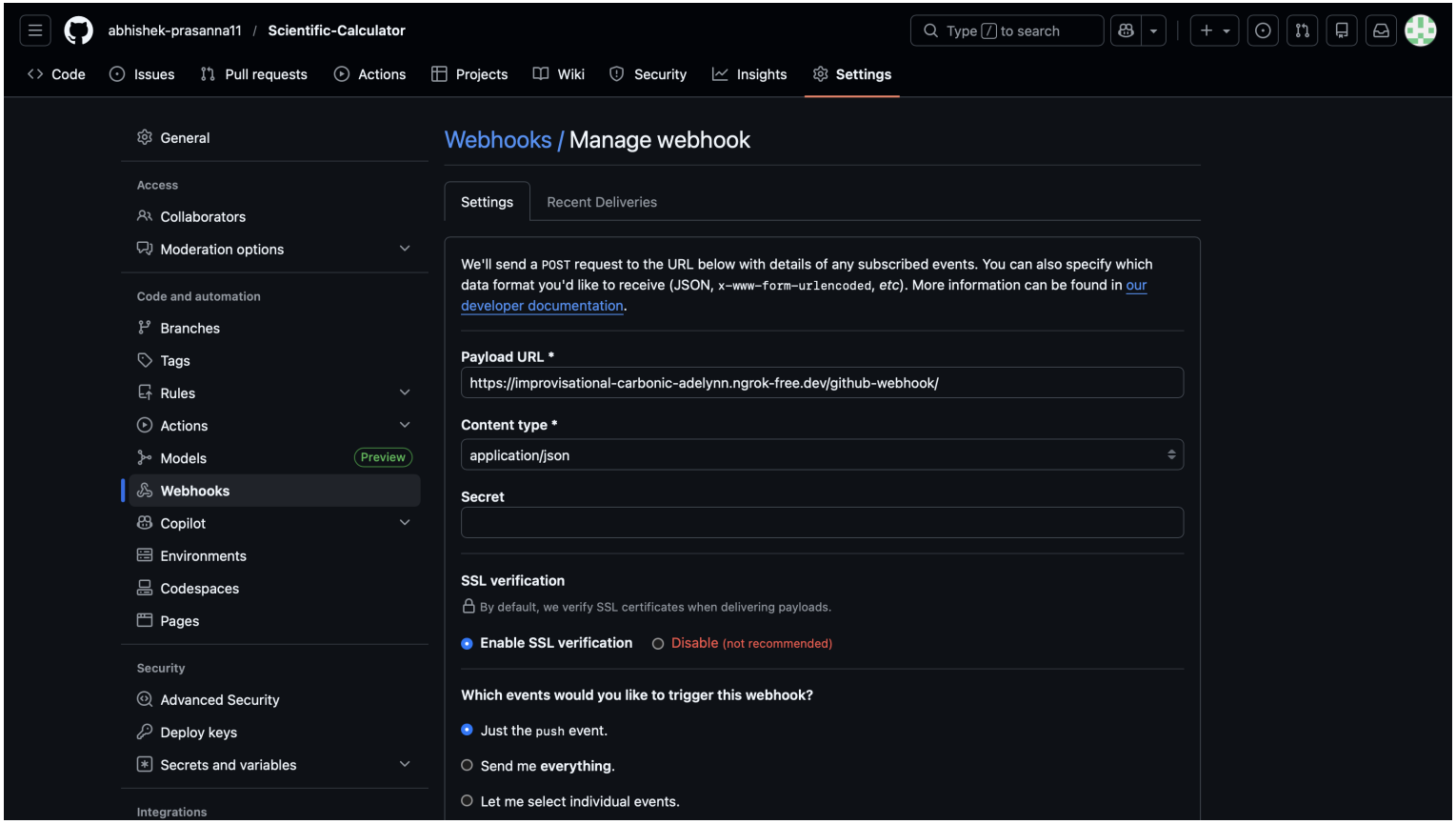
- Uploads the Docker image to Docker Hub
- Enables centralized storage of container images
- Allows deployment systems to pull the latest version

Deployment Stage
In the final stage, Jenkins triggers an Ansible playbook to deploy the latest application container.

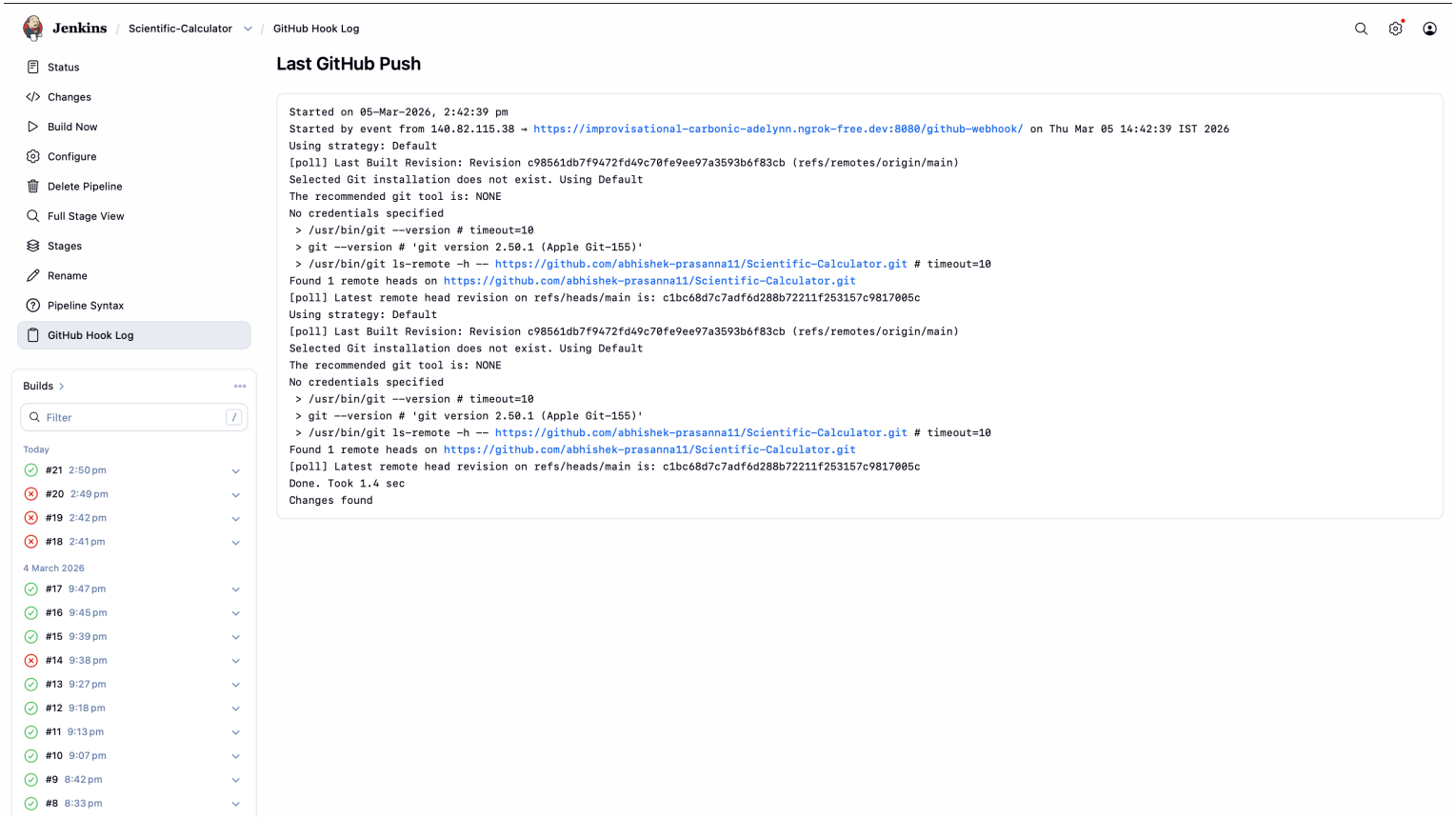
- Automates application deployment
- Ensures the latest version of the application is deployed
- Reduces manual intervention in the deployment process

Jenkins Pipeline Execution

GitHub Webhook Configuration



GitHub Webhook Logs



Ngrok Tunnel

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ngrok                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
(Ctrl+C to quit)

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Session Status                online
Account                       abhishek1109prasanna@gmail.com (Plan: Free)
Update                        update available (version 3.36.1, Ctrl-U to update)
Version                       3.35.0
Region                        India (in)
Latency                       24ms
Web Interface                 http://127.0.0.1:4040
Forwarding                    https://improvisational-carbonic-adelynn.ngrok-free.dev -> http://localhost:8080

Connections                   ttl     opn     rt1     rt5     p50     p90
                              35      0       0.00    0.00    30.07   31.52

HTTP Requests
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14:42:39.161 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
14:41:34.021 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
21:46:57.762 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
21:43:40.753 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
21:43:36.313 IST OPTIONS /administrativeMonitor/hudson.diagnosis.ReverseProxySetupMonitor/testForReverseProxySetup/http://localhost:8080/manage// 403 Forbidden
21:42:13.714 IST OPTIONS /administrativeMonitor/hudson.diagnosis.ReverseProxySetupMonitor/testForReverseProxySetup/http://localhost:8080/manage// 403 Forbidden
21:39:24.600 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
21:38:27.087 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
21:27:04.577 IST POST    /github-webhook/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
21:23:25.767 IST OPTIONS /administrativeMonitor/hudson.diagnosis.ReverseProxySetupMonitor/testForReverseProxySetup/http://localhost:8080/manage// 403 Forbidden
```

Dockerfile Explanation

The Dockerfile is a configuration file used to build Docker container images. It defines the instructions required to assemble the runtime environment for the application.
Docker allows applications to run inside isolated environments called containers. These containers include all necessary dependencies, ensuring consistent behavior across different machines.

Base Image

The base image provides the runtime environment for the container. In this project, a Java runtime image is used as the base since the application is written in Java.

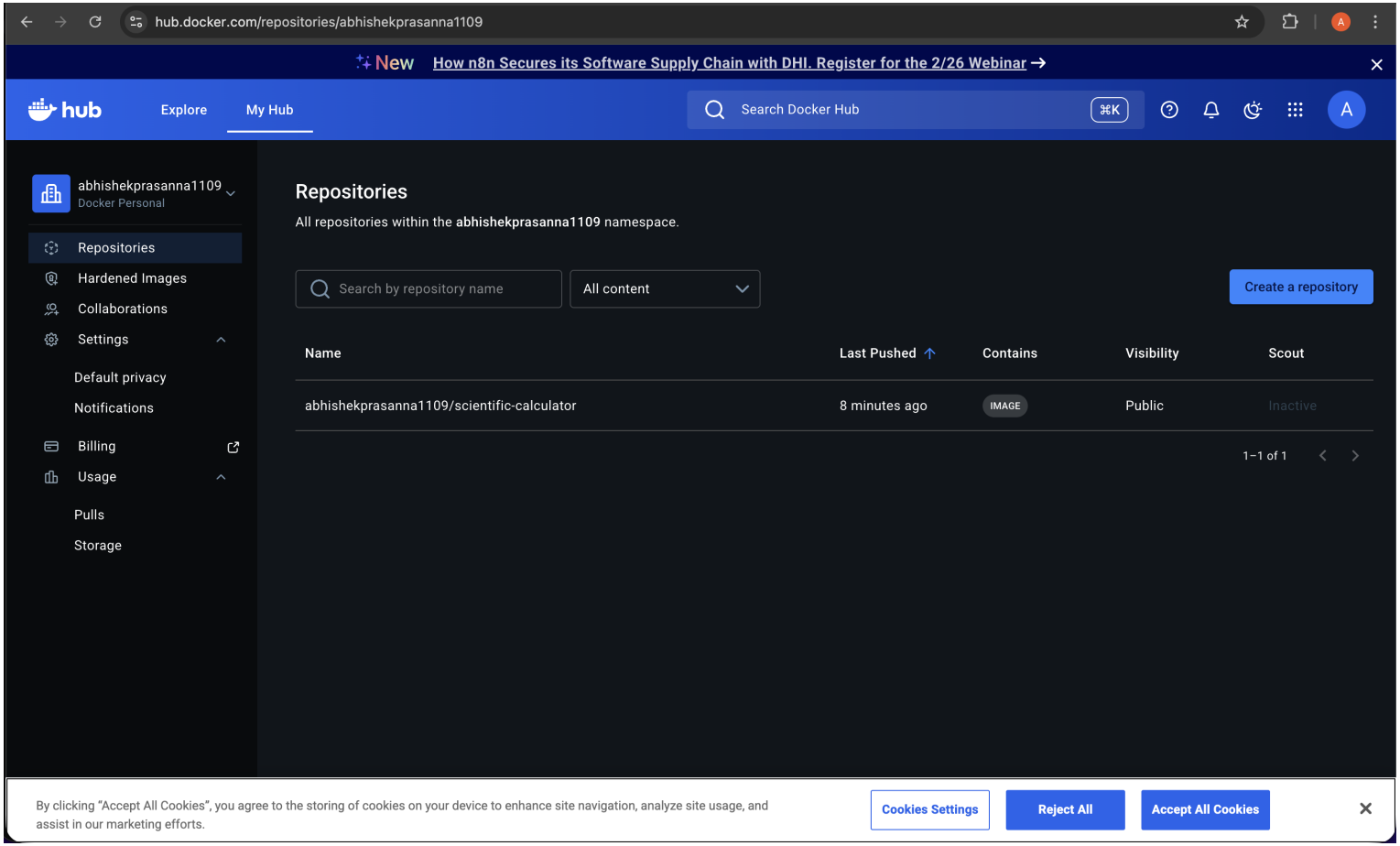
Copying Application Files

The compiled JAR file generated during the Maven build process is copied into the Docker image.

Application Execution

The Dockerfile defines the command that should run when the container starts.

Docker Hub Repository



Ansible Deployment

Ansible is an infrastructure automation tool used to automate deployment tasks. In this project, an Ansible playbook named `deploy.yml` is used to automate the deployment of the application container.

Playbook

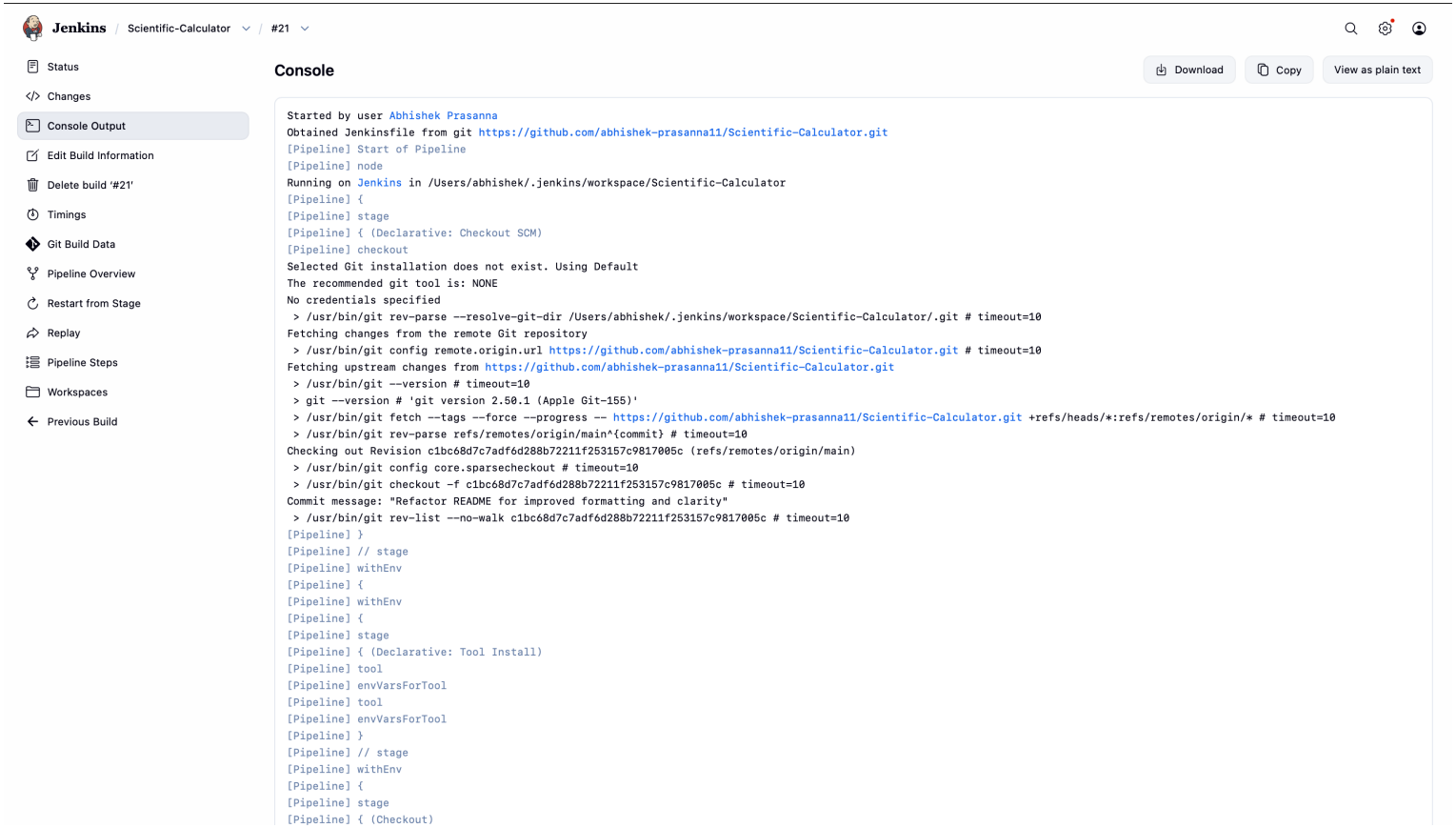
A playbook is a YAML file that defines a set of automation tasks to be executed on target systems.

- Target hosts
- Tasks to be executed
- Order of execution

Deployment Tasks

- Pull the latest Docker image from Docker Hub
- Stop any existing container running the application
- Start a new container using the updated image

Application Console Output



Sample Application Output

```
Scientific-Calculator — -zsh — 207×62

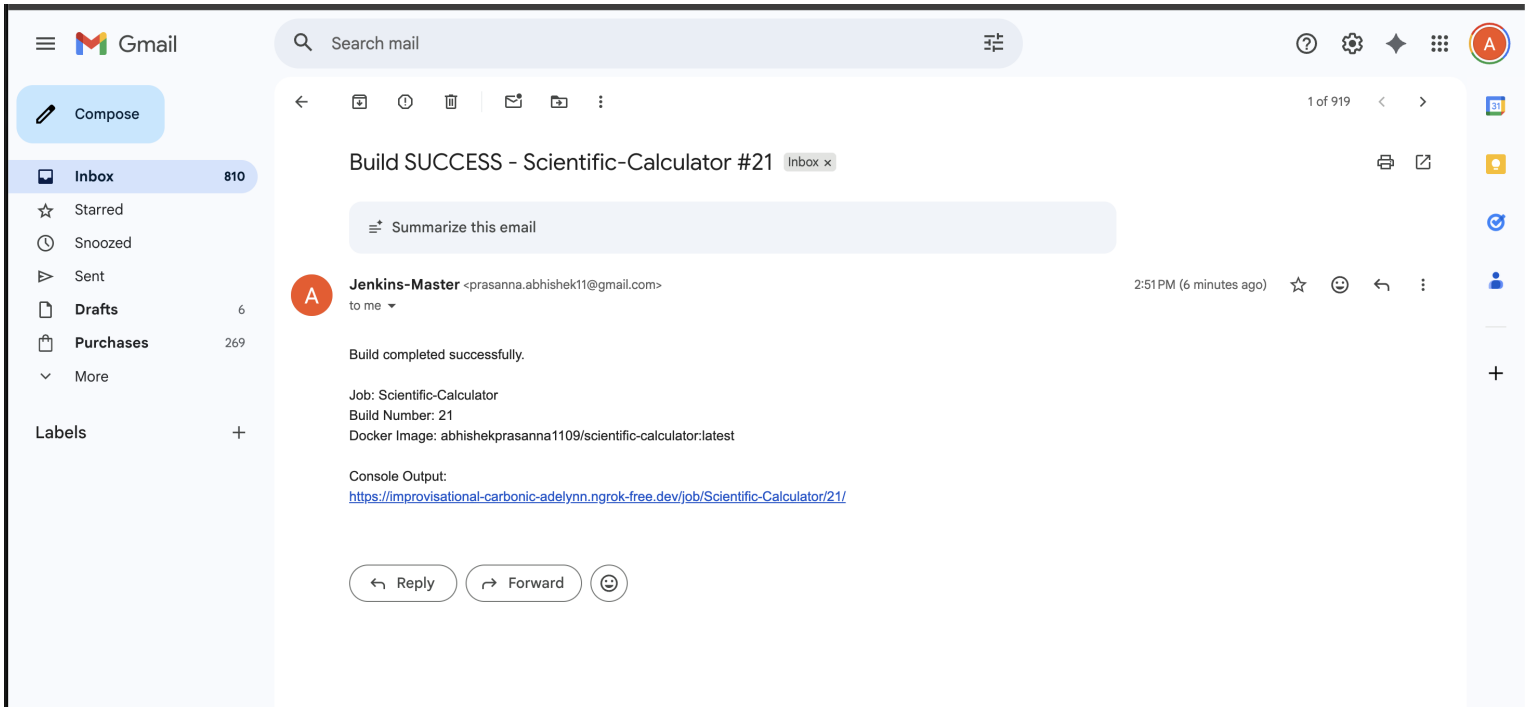
abhishek@Abhisheks-MacBook-Air Scientific-Calculator % docker images

IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
abhishekprasanna1109/scientific-calculator:latest   14eacf88a6ec       667MB          197MB    U
bill:latest                                    79cb4dad9c56       784MB          247MB
ing-service:latest                             79cb4dad9c56       784MB          247MB
abhishek@Abhisheks-MacBook-Air Scientific-Calculator % docker ps
CONTAINER ID   IMAGE                                          COMMAND                  CREATED              STATUS              PORTS     NAMES
49a4babe359e   abhishekprasanna1109/scientific-calculator:latest   "/__cacert_entrypoin…"   About a minute ago   Up About a minute             scientific-calculator
abhishek@Abhisheks-MacBook-Air Scientific-Calculator % docker exec -it scientific-calculator java -jar app.jar


Scientific Calculator
1. Add
2. Subtract
3. Multiply
4. Divide
5. Square Root
6. Factorial
7. Natural Log
8. Power
9. Exit
Choose option: 8
Enter base: 2
Enter exponent: 3
Result: 8.0

Scientific Calculator
1. Add
2. Subtract
3. Multiply
4. Divide
5. Square Root
6. Factorial
7. Natural Log
8. Power
9. Exit
Choose option: 9
Exiting...
abhishek@Abhisheks-MacBook-Air Scientific-Calculator %
```

Email Notification



Conclusion

- Successfully implemented a full DevOps CI/CD pipeline
- Integrated GitHub, Jenkins, Docker, and Ansible
- Achieved automated build, test, and deployment
- Demonstrated a real-world DevOps workflow